Literature Survey for Mine Proximity Safety Device (Team A07)

Our device puts RFID tags on workers and a reader on each machine, so both get warned before a collision. It will be piloted at SCCL's RG-1 opencast area. We kept 16 papers: 6 Good, 6 Decent and 4 Just in case. Our main base paper is the NIOSH RFID collision-avoidance study (Ruff & Hession-Kunz, 2001), with LANDMARC (2003) for locating the tag.

At a glance

Papers 1–6 shape the device itself; papers 7–16 support testing, machine learning and background. Change a paper's tier from the dropdown if the team moves it.

#	Paper	Year	Published in	Tier	What we use it for
1	Application of Radio-Frequency Identification Systems to Collision Avoidance in Metal/Nonmetal Mines	2001	IEEE Trans. Industry Applications	Good	Base paper: worker tag + machine reader, device specs
2	LANDMARC: Indoor Location Sensing Using Active RFID	2003	IEEE PerCom	Good	Locating a tag from signal strength (nearest-neighbour method)
3	Recommendations for Evaluating & Implementing Proximity Warning Systems on Surface Mining Equipment	2007	NIOSH Report of Investigations 9672	Good	Field-test procedure and tag-system results on mine trucks
4	Event-Based Sensor Data Scheduling: Trade-Off Between Communication Rate and Estimation Quality	2013	IEEE Trans. Automatic Control	Good	Tag update rate vs battery life
5	NLOS Identification and Mitigation for Localization Based on UWB Experimental Data	2010	IEEE J. Selected Areas in Communications	Good	Spotting and correcting blocked signals with ML (SVM)

#	Paper	Year	Published in	Tier	What we use it for
6	Robust Acoustic Positioning for Safety Applications in Underground Mining	2015	IEEE Trans. Instrumentation and Measurement	Good	Worker position and speed relative to a machine
7	Method for Testing Proximity Detection and Alert Technology for Safe Construction Equipment Operation	2013	Construction Management and Economics	Decent	Test method for a 434 MHz tag–reader alarm system
8	Evaluation of a Radar-Based Proximity Warning System for Off-Highway Dump Trucks	2006	Accident Analysis & Prevention	Decent	Evidence that radar alone gives nuisance alarms
9	Application of Ultra-Wide Band Sensors in Mining	2023	MDPI Sensors	Decent	UWB tags tested inside a real mine
10	Improving Indoor Localization Using Convolutional Neural Networks on Computationally Restricted Devices	2018	IEEE Access	Decent	Deep-learning version of paper 5 on low-power hardware
11	Human Detection and Activity Classification Based on Micro-Doppler Signatures Using Deep Convolutional Neural Networks	2016	IEEE Geoscience and Remote Sensing Letters	Decent	Radar telling a person apart, for workers without tags
12	Determining Proximity Warning and Action Zones for a Magnetic Proximity Detection System	2011	IEEE IAS Annual Meeting	Decent	Warning zones and machine shutdown
13	Opportunities for Sensing Systems in Mining	2017	IEEE Trans. Industrial Informatics	Just in case	Survey of mining sensors for the literature review
14	Collision Risk-Minimized Multivehicles Cooperative Path Planning in Open-Pit Mine Unloading Areas	2026	IEEE Sensors Journal	Just in case	Open-pit background for the introduction

#	Paper	Year	Published in	Tier	What we use it for
15	Seeing Through Fog Without Seeing Fog: Deep Multimodal Sensor Fusion in Unseen Adverse Weather	2020	IEEE/CVF CVPR	Just in case	Idea of trusting whichever sensor is not blinded
16	Underground Mine Positioning: A Review	2022	IEEE Sensors Journal	Just in case	Review of positioning methods used in mines

Papers 3, 7, 8 and 9 are not IEEE; their publishers are linked.

Good papers

These six papers define the device: the tag–reader design, how to locate the tag, how often it transmits, and how to test it.

1. Application of Radio-Frequency Identification Systems to Collision Avoidance in Metal/Nonmetal Mines

Ruff & Hession-Kunz · IEEE Trans. Industry Applications 37(1), 2001 · IEEE · free NIOSH copy

Paper. NIOSH's study of radio tags that warn equipment operators about workers in blind spots. It is our base paper because it is the same device: a tag on the worker, a reader on the machine.

Why we selected it as Good

- It is the same device as ours: a tag on the worker, a reader on the machine and an alarm in the cab, tested on mining equipment.
- It is the only paper that is IEEE, about mining and about RFID all at once, so it is our main base paper.
- Its open problems (tag orientation, signal nulls, no position) are exactly what our project improves.

Implementation

- Set the specs first: 50 ft range, at most 0.5 s delay, 12 h battery, tag under $40, reader under $1,200, low RF power for blasting caps and human exposure.

- Tested two off-the-shelf systems on a front-end loader. UHF active tags read only every 6 s and were reliable within about 20 ft; reader $5,000, tag $100.
- LF passive tags met the cost target but reached only about 7 ft, and the tags were too directional.
- Built a custom prototype with ID International: a hard-hat tag with a microprocessor, 315 MHz transmitter and coin-cell battery, sending "I'm here" 3–4 times a second.
- The reader (receiver, signal conditioning, microprocessor, whip antenna, 12 V supply) drives a cab alarm with a light and 105 dB buzzer. A "double buzz" flags two or more tags, up to four.
- Mounted the reader on the loader's rear at about 7 ft and mapped reliable, upright-only and sporadic detection zones.

Limitations

- Tags were missed when the worker bent over, and in signal nulls around the vehicle.
- A sporadic zone at the edge of range caused false alarms from tags far away.
- It only tells whether a tag is in range, not where the worker is.
- Prototype stage, 2001 hardware; RF exposure and blasting-cap safety tests were still pending.

What we take: the tag–reader architecture and its spec table, updated for SCCL haul trucks.

2. LANDMARC: Indoor Location Sensing Using Active RFID

Ni, Liu, Lau & Patil · IEEE PerCom 2003 · 2,323 citations (OpenAlex) · IEEE · free copy

Paper. Locates active RFID tags by comparing their signal strength with reference tags placed at known spots.

Why we selected it as Good

- The most cited RFID locating paper we found: 2,323 citations, IEEE.
- It turns reader signal strength into a worker position with k-nearest neighbours, which gives our Communications + ML project its ML part.
- Its slow updates (about a minute per fix) are a clear weakness we can fix and measure against.

Implementation

- RF Code Spider readers and active tags at 308 MHz. Readers report 8 power levels; tags beacon every 7.5 s on average.
- 4 readers, 16 reference tags in a grid about 1 m apart, 8 tracking tags, 48 hours of data.
- For each tracking tag, compute its signal-strength distance E to every reference tag across all readers.
- Take the k nearest reference tags and average their positions, weighted by $1/E^2$.
- Best k = 4: average error 1.09 m, worst 1.81 m. With 3 readers the worst error grew to 2.59 m.

Limitations

- About one minute to scan all 8 power levels, plus the 7.5 s beacon gap: far too slow for a moving machine.
- Tags at the same spot reported different power levels.
- People and furniture caused large errors; indoor, static tags only.
- Needs fixed reference tags (about one per m^2), which a mine site would have to mount on the machine or roadside.

What we take: the k-nearest-neighbour locating method, our simplest ML baseline for tag position.

3. Recommendations for Evaluating & Implementing Proximity Warning Systems on Surface Mining Equipment

T. Ruff · NIOSH Report of Investigations 9672, 2007 · free NIOSH copy

Paper. NIOSH's field guide to choosing and testing proximity warning systems on surface-mine equipment.

Why we selected it as Good

- NIOSH's own guide for proximity warning on surface-mine trucks, the setting closest to SCCL's opencast mines.
- It tests RF and magnetic tag systems as well as radar, so it supports our choice of tags.
- It gives the test procedure and detection-zone definitions we will use in our pilot.
- We moved it up from Decent once the team chose the RFID route.

Implementation

- Plotted the blind areas of five types of surface mining equipment.

- Tested radar, sonar, GPS, radio tags and radar–camera combinations on off-highway dump trucks, mainly at the Phelps Dodge Morenci copper mine, Arizona.
- Tag systems tested: AMT CAS-CAM/RF (RF units on Caterpillar 793B haul trucks, tags on light vehicles), HASARD and Nautilus Buddy.
- HASARD: a magnetic field from the machine, a 3-axis receiver in the worker's tag, and two-way RF so both ends alarm; about 4.6 m radius.
- Gives a step-by-step test: a person and a car approach on a 1 m grid. Reliable zone = detected 100% of the time; sporadic = about 10–100%.

Key findings

- Radar and sonar alarm on rocks, berms and ruts. Tag systems alarm only for tagged objects, so they miss anyone without a tag.
- Ideally the detection range should grow with the truck's speed.
- Automatic braking from sensor alarms is not recommended because of nuisance alarms. Pair alarms with cameras.

Limitations

- 2007 products; no wearable worker tags existed for the AMT system at test time.
- Tag-system tests were short because tagging every worker was too costly.
- It did not measure whether accidents actually fell.
- US surface mines only; no ML.

What we take: the test procedure for our SCCL pilot, and the idea of a speed-dependent warning range.

4. Event-Based Sensor Data Scheduling: Trade-Off Between Communication Rate and Estimation Quality

Wu, Jia, Johansson & Shi · IEEE Trans. Automatic Control 58(4), 2013 · IEEE

Paper. Shows how a battery-powered sensor can send fewer updates while the receiver still tracks it well.

Why we selected it as Good

- It is the only paper behind our claim "faster updates when danger is close, saves battery when it isn't".
- Its formula linking the send threshold to the send rate lets us estimate tag battery life.

- Published in a top journal, IEEE Transactions on Automatic Control; it was in our original Good list.

Implementation

- A sensor measures a linear system and decides at each step whether to transmit.
- It sends only when its "surprise" (the normalised gap between its reading and the receiver's prediction) exceeds a threshold δ.
- When nothing arrives, the receiver still learns that the surprise was small and uses that in its estimate.
- Derives the exact estimator (too heavy to run), a Kalman-filter-like approximation, and a formula linking δ to the average send rate.
- Simulation with two sensors sharing one channel: error for one process fell 52.7% while the other rose only 5.83%.

Limitations

- Theory and simulation only; no hardware or battery measurements.
- Assumes linear Gaussian models and a feedback link from the receiver to the sensor.
- One fixed threshold; it does not change δ as danger grows.

What we take: the rule for our tag: transmit rarely when far and steady, more often near the machine by lowering δ.

5. NLOS Identification and Mitigation for Localization Based on UWB Experimental Data

Maranò, Gifford, Wymeersch & Win · IEEE J. Selected Areas in Communications 28(7), 2010 · IEEE

Paper. Uses machine learning to tell whether a radio signal was blocked, and to correct the distance error that causes.

Why we selected it as Good

- It uses ML to fix the biggest radio problem in a mine: signals blocked by bodies, rock or machines.
- It fits our Communications + ML domain exactly, and is well cited (668 citations, IEEE JSAC).
- It was in our original Good list.

Implementation

- Measured at MIT in 2007 with two off-the-shelf UWB radios (3.1–6.3 GHz) on carts at 90 cm, at over 100 points 0.6–18 m apart.
- Offices, hallways, a lab and a lobby; 1,024 signals, half clear line of sight and half blocked.
- Six features per received pulse: energy, peak amplitude, rise time, mean excess delay, RMS delay spread, kurtosis.
- An LS-SVM classifier labels each signal clear or blocked; an LS-SVM regressor predicts and subtracts the distance error.
- Over 91% correct classification using three features (energy, rise time, kurtosis).
- For blocked signals, ranges within 1 m rose from about 30% to 60% after correction. In 5-anchor simulations, identify-and-correct beat ignoring blockage.

Limitations

- Needs the full received pulse, sampled every 41 ps; ordinary RFID readers give only signal strength.
- Indoor offices, static radios; no mine and no moving machine.
- Trained in one building, so a new site may need retraining.

What we take: the identify-then-correct idea, applied to our reader's readings when a body or machine blocks the tag.

6. Robust Acoustic Positioning for Safety Applications in Underground Mining

Pfeil, Pichler, Schuster & Hammer · IEEE Trans. Instrumentation and Measurement 64(11), 2015 · IEEE

Paper. Locates a miner around an underground machine using sound, with a microphone tag worn by the miner.

Why we selected it as Good

- Same problem as ours (a worker crushed near a mining machine) and the same layout: units on the machine, a tag on the worker.
- It gives the worker's position and speed, which our warning levels need.
- A full IEEE Transactions paper with tests beside a real mining machine, and one of the team's starred papers.

Implementation

- Loudspeakers on the machine face the access area; the miner wears a transponder with a microphone.
- A master unit sends an RF trigger to sync timing, then the speakers send chirps at 5–20 kHz.
- Time of flight from at least two speakers gives the 2-D position. Up- and down-chirps also give the miner's speed and direction.
- Hyperbolic chirps are unaffected by the Doppler shift of a walking miner; detection works down to −30 dB signal-to-noise ratio.
- Lab grid errors were below 3 cm. Beside a real mining machine in metal lining, machine noise (mostly below 5 kHz) had no effect.
- Moving tests used a linear rail at up to 1 m/s.

Limitations

- Not tested in a real coal mine: parts would need ATEX explosion-safety certification.
- Needs line of sight; blocked paths must be thrown away.
- Chirps of up to 500 ms make each update slow.
- In tests the microphone was cabled to the audio interface, not wireless.
- Open-pit wind, rain and long distances are untested.

What we take: using the RF link for timing, and measuring how fast the worker closes on the machine, which our alert levels need.

Decent papers

These six papers back up the design: how to test alarms, why radar alone fails, and the ML and shutdown pieces.

7. Method for Testing Proximity Detection and Alert Technology for Safe Construction Equipment Operation

Marks & Teizer · Construction Management and Economics 31(6), 2013 · Taylor & Francis · free related PhD thesis

Paper. A step-by-step method for testing whether a worker-tag alarm system warns in time. Its hardware is the closest match to our device.

Why we selected it as Decent

- Its hardware is the closest to our device (cab unit plus worker tag at 434 MHz, alarm at both ends), and its test method is easy to copy.
- Not Good because it covers construction sites, not mines, is not IEEE, and shows how to test a system rather than design one.
- The team placed it in Decent after weighing these points.

Implementation

- Prototype RF system at about 434 MHz. A unit in the cab (antenna, reader, alarm) broadcasts; a battery tag on the worker's hard hat picks it up within a set distance.
- When the tag hears the signal, both the worker and the operator are alarmed at once.
- Stage 1, machine parked: a worker walked in from 36 evenly spaced angles, twice each, with short, medium and long-range tags. Nine machines, from dump truck to excavator.
- On an asphalt paver, 99.5% of 432 approaches triggered an alert; median alert distances were 13.8, 14.8 and 16.9 m for the three tags.
- Stage 2, machine moving: a pickup at 16 km/h drove at a standing worker. All 96 runs alerted, with no false alarms.
- Live site trial at a coal power plant project: about 200 alerts, from 2.8 m to 62.5 m.

Limitations

- Alert distance varied widely: the short-range tag fired anywhere from 0 to 18.5 m on the paver.
- Only says "in range" or not; no position or closing speed.
- Construction sites on flat paved ground, not mines; only one moving speed tested.
- Prototype internals (circuit, signal design) are not described. Not an IEEE paper.

What we take: the test protocol (36 angles, three ranges, moving-machine runs) to copy at SCCL.

8. Evaluation of a Radar-Based Proximity Warning System for Off-Highway Dump Trucks

T. Ruff · Accident Analysis & Prevention 38(1), 2006 · Elsevier · free NIOSH copy

Paper. Two years of NIOSH tests of a radar blind-spot alarm on a large haul truck.

Why we selected it as Decent

- Real haul-truck alarm data (82% true, 18% false, 42% nuisance) that backs our choice of tags over radar alone.

- Not Good because it is about radar, which is not our main sensor, and it is not an IEEE paper.

Implementation

- Preco PreView pulsed radar on a Caterpillar 793B (260 t) truck at the Phelps Dodge Morenci copper mine, Arizona.
- Two antennas on the rear axle (angled 15° inward) and two on the front bumper, plus front and rear cameras with a cab monitor.
- A standing person was detected up to 7.6 m from the rear axle.
- Rear alarms and video were recorded for 7 days of normal work: 618 reversing moves, 580 alarms.
- Results: 82% true alarms, 18% false, 42% nuisance (real object, no danger), and no missed detections.

Limitations

- Radar cannot tell a person from a berm, rock or parked machine, hence the nuisance alarms.
- Tuned to ignore the tyres, it could miss a person within 3 m of the rear axle or under the beam.
- Rear data only; one truck and one mine.

What we take: the evidence for using worker tags instead of radar alone, and the 82/18/42% figures to beat.

9. Application of Ultra-Wide Band Sensors in Mining

Wisiak, Jakić & Hartlieb · MDPI Sensors 23(1), 2023 · MDPI (free)

Paper. A short perspective piece testing a cheap UWB tag kit inside an Austrian test mine.

Why we selected it as Decent

- The only paper testing low-cost radio tags inside a real mine; recent (2023) and one of the team's starred papers.
- It suggests linking tags to an emergency machine shutdown, which supports our no-start feature.
- Not Good because it is a six-page perspective with no accuracy figures or ML, and its range was only 28 m.

Implementation

- Qorvo/Decawave MDEK1001 kit (DWM1001C modules, two-way ranging), an Android tablet app and CR123 batteries.
- Four anchors fixed to notches in the drift wall, in a rectangle up to 4 m wide, on average more than 1.2 m above the target.
- The moving tag rode on a remote-controlled car.
- Findings: 28 m maximum between two sensors without signal loss; accurate only with clear line of sight.
- With several tags moving, one tag shadowed another and dropped off the map.
- No interference with the mine's WLAN, radio system, Bluetooth or GPS.

Limitations

- Six-page perspective with no accuracy figures and no ML.
- Tiny test area; anchors on walls, not on machines; an RC car, not a real worker or vehicle.
- 28 m is short for a loaded dumper at speed.

What we take: proof that UWB tags work underground, and its suggestion to link tags to an emergency machine shutdown.

10. Improving Indoor Localization Using Convolutional Neural Networks on Computationally Restricted Devices

Bregar & Mohorčič · IEEE Access 6, 2018 · IEEE (free, open access)

Paper. The deep-learning version of paper 5: CNNs fix blocked-signal errors, small enough for low-power hardware.

Why we selected it as Decent

- The deep-learning upgrade of paper 5, with public data and code, so we can train a model before our hardware is ready.
- Not Good because it needs UWB pulse data that ordinary RFID readers do not give, and it was tested indoors only.
- We moved it down from Good after the team chose RFID.

Implementation

- Input is the raw UWB channel impulse response, with no hand-made features.
- One CNN classifies each signal as clear or blocked; a second predicts the distance error.

- The predictions feed least-squares and weighted least-squares position solvers.
- Runtime was compared across computers from powerful to very limited.
- Code (TensorFlow) and the authors' dataset are public: 42,000 samples from 7 indoor sites, recorded with Decawave DWM1000 radios.

Limitations

- Needs UWB pulse data; plain RFID readers cannot supply it.
- Indoor offices, static tags; no people moving, no machines.
- The abstract gives no accuracy figures; check the full paper before quoting numbers.

What we take: a ready dataset and code to train our first blocked-signal model before we have hardware.

11. Human Detection and Activity Classification Based on Micro-Doppler Signatures Using Deep Convolutional Neural Networks

Kim & Moon · IEEE Geoscience and Remote Sensing Letters 13(1), 2016 · IEEE

Paper. Teaches a CNN to recognise people from the way their limbs move in radar returns.

Why we selected it as Decent

- A well-cited (654 citations) way for radar and ML to recognise a person, which backs our "person versus rock or vehicle" claim.
- Not Good because it covers workers without tags, now secondary to our RFID focus, and its radar was static with no position output.
- We moved it down from Good after the team chose RFID.

Implementation

- A Doppler radar records micro-Doppler: small frequency shifts from swinging arms and legs.
- The CNN reads the raw spectrogram directly and learns its own features.
- 97.6% accuracy at detecting humans and 90.9% at classifying activities.

Limitations

- A short letter with few details on the data.
- Radar stood still; a moving, vibrating dumper adds its own Doppler.
- Gives no distance or position; one target at a time.
- Only covers workers without tags, which is now secondary for us.

What we take: the fallback for detecting a person who is not wearing a tag.

12. Determining Proximity Warning and Action Zones for a Magnetic Proximity Detection System

Jobes, Carr, DuCarme & Patts · IEEE IAS Annual Meeting 2011 · IEEE · free NIOSH copy

Paper. NIOSH software that turns a worker's tag position into zones that stop only the dangerous machine motions.

Why we selected it as Decent

- The only paper that links a worker's position to stopping specific machine functions, which is our "no start during repairs" feature.
- Not Good because it is underground and magnetic, has few citations, and was not yet tested for accuracy on a working machine.

Implementation

- Generators on a remote-controlled continuous mining machine create a magnetic field; the worker's Personal Alarm Device senses it.
- The device's position is found to within 20–50 cm and treated as a circle of uncertainty.
- Ten zones around the machine, each tied to the functions it blocks: tramming forward or back, conveyor swing, cutter-head movement and so on.
- If the circle touches a zone, only that zone's functions are disabled, instead of shutting down the whole machine.

Limitations

- Accuracy on a working machine was not yet measured.
- Underground continuous miners only; magnetic fields have short range.
- The zones shown are one example, not a tested recommendation.

What we take: the zone-to-function idea for our "no start during repairs" feature.

Just in case papers

These four give background for the introduction and literature review; none of them shapes the device.

13. Opportunities for Sensing Systems in Mining

Kiziroglou, Boyle, Yeatman & Cilliers · IEEE Trans. Industrial Informatics 13(1), 2017 · IEEE

Paper. A survey of the sensors used in mining today and those likely to pay off within ten years.

Why we selected it as Just in case

- An IEEE survey with citable facts for our literature review, such as RFID's short range and hazard-area fencing with shutdown.
- Only Just in case because it is a survey, with no system or tests of its own.

Implementation (survey content)

- Compares localization options in a table: RFID reaches about 10 m, is cheap and simple, but gives no range information.
- Radar, sonic and optical sensors reach under 100 m and measure distance and speed, but are sensitive to the environment.
- Describes "hazard-area fencing": wearable detectors that warn and trigger a preventive shutdown, citing NIOSH's HASARD.
- Predicts energy-harvesting, self-powered sensor nodes as a key next step.

Limitations: a survey with no new system or tests; proximity detection gets only a few paragraphs.

What we take: citable figures for why plain RFID needs extra ranging, and support for tags that cut machine power.

14. Collision Risk-Minimized Multivehicles Cooperative Path Planning in Open-Pit Mine Unloading Areas

Chen, Zhu, Ke, Yang, Yang & Ge · IEEE Sensors Journal 26(3), 2026 · IEEE

Paper. Plans paths for several driverless dump trucks in an open-pit unloading area so they do not collide.

Why we selected it as Just in case

- The only paper set in an open-pit mine, and the newest on our list (2026); one of the team's starred papers.
- Only Just in case because it plans paths for driverless trucks: no people, no tags, and it assumes positions are known exactly.

Implementation

- An improved Hybrid A* planner with dynamic routing potential fields spreads the trucks' paths apart.
- A second optimiser smooths each path and keeps a safety distance between trucks.
- MATLAB simulations of the Haluusu open-pit mine, Inner Mongolia, with 2–5 trucks.
- For two trucks, the collision-free share of the path rose from 42.23% to 64.17%; the full method reached 100% collision-free paths.

Limitations

- Trucks versus trucks only; no people.
- Assumes exact localization with no sensors, a known map and no communication delays.
- Simulation only.

What we take: recent open-pit context for the introduction.

15. Seeing Through Fog Without Seeing Fog: Deep Multimodal Sensor Fusion in Unseen Adverse Weather

Bijelic et al. · IEEE/CVF CVPR 2020 · IEEE · free PDF

Paper. A deep network that fuses several sensors and leans on whichever one the weather has not blinded.

Why we selected it as Just in case

- A well-cited (609 citations) example of combining sensors when weather blinds some of them, which backs our dust, rain and night claim.
- Only Just in case because its sensors are too costly for our budget, and it is built for cars, not mines or RFID.

Implementation

- Car sensors: camera, lidar, radar, gated near-infrared camera and thermal camera.
- Fusion weights adapt to how much information each sensor carries at that moment.
- Trained on clear-weather data only, then tested in fog, snow and rain.
- Dataset: over 10,000 km of driving in northern Europe with 100,000 labelled objects, plus fog-chamber recordings.

Limitations: lidar and gated cameras are costly, it needs heavy GPU compute, and it was built for cars on roads, not dust.

What we take: the idea of trusting the sensor that still works when dust or rain blinds another.

16. Underground Mine Positioning: A Review

Seguel et al. · IEEE Sensors Journal 22(6), 2022 · IEEE

Paper. Reviews positioning technologies for underground mines against what users need from them.

Why we selected it as Just in case

- A recent IEEE review of positioning in mines, useful for the literature survey and more relevant if the pilot moves underground.
- Only Just in case because it is a review with no method of its own, covers underground rather than open-pit mines, and we have read only its abstract.

Implementation (review content)

- Starts from the 2006 US rule (Public Law 109-236) requiring mines to know each worker's location.
- Surveys positioning technologies and judges existing systems for underground tracking.
- Ends with future research directions.

Limitations: a review with no method or results of its own; underground, not open pit. We have read only its abstract so far.

What we take: a recent IEEE source for the literature review, and stronger relevance if our pilot moves underground.

Gaps our device fills

No paper tests worker RFID tags on open-pit haul trucks at speed. That gap and seven others are what our project adds.

Gap	Closest paper(s)	What our device adds
Worker tags on open-pit haul trucks at speed	1 (loader prototype), 3 (tags on light vehicles only), 7 (construction, 16 km/h)	Worker tags and readers on SCCL dumpers, tested at haul speeds

Gap	Closest paper(s)	What our device adds
Warning range that grows with truck speed	3 recommends it; nobody built it	Alert distance set from speed and stopping distance
Update rate that rises with danger	4 (fixed threshold, simulation only)	Tag sends more often in closer zones, saves battery when far
ML to correct blocked readings	5 and 10 (UWB pulse data, indoor offices)	ML on our reader's signal readings with bodies, rocks and trucks in the way
Telling a person from a rock or vehicle	8 (radar cannot), 11 (radar CNN in a lab)	Tag ID confirms a person; radar with ML as fallback for untagged workers
Dust, rain and night in an open pit	15 (fog and snow on roads)	Pilot at SCCL RG-1 opencast in real conditions
No-start lock during repairs	12 (zones disable functions, underground, untested), 9 (suggests shutdown)	A tag inside the repair zone blocks engine start
Indian mine conditions	None	First trial at SCCL RG-1

Blocking a start while someone is in the repair zone is different from automatic braking, which paper 3 advises against for moving trucks.

Papers not selected

We dropped eight papers. The first two are papers 1 and 2 from our first set, the car-steering studies.

Paper	Year	Published in	Why we dropped it
Modeling Lateral Control Behavior of Driver for Manual Takeover of Automated Vehicles During Pedestrian Collision Avoidance	2024	IEEE Trans. Vehicular Technology	Models a driver's steering in a car simulator; no sensing, no mining

Paper	Year	Published in	Why we dropped it
Efficient Deep Reinforcement Learning With Expert Demonstrations for Human–Machine Shared Steering Control Under Emergency Obstacle Avoidance Conditions	2026	IEEE Trans. Vehicular Technology	Shared steering for cars; assumes the obstacle is already known
Localization via Ultra-Wideband Radios: A Look at Positioning Aspects for Future Sensor Networks	2005	IEEE Signal Processing Magazine	General tutorial; papers 5 and 10 cover the theory we need
UWB NLOS/LOS Classification Using Deep Learning Method	2020	IEEE Communications Letters	Does a smaller part of what paper 10 does
A Deep Learning-based Radar and Camera Sensor Fusion Architecture for Object Detection	2019	IEEE SDF Conference	Camera detection for cars; paper 15 covers the fusion idea
Human Activity Classification Based on Micro-Doppler Signatures Using a Support Vector Machine	2009	IEEE Trans. Geoscience and Remote Sensing	Replaced by the same group's CNN version, paper 11
Multispectral Pedestrian Detection: Benchmark Dataset and Baseline	2015	IEEE CVPR	Cameras only, city streets
Real-Time People Tracking and Identification From Sparse mm-Wave Radar Point-Clouds	2021	IEEE Access	Indoor radar with no tags; dropped once we chose the RFID route

Sources

Every title in the At a glance table links to its publisher page. Download the IEEE papers through the NIT Warangal IEEE Xplore login.

Details for papers 1, 3–9, 13 and 14 come from their full PDFs; 2 and 12 from free full copies; 15 from the authors' project page; 10, 11 and 16 from abstracts only. Check those three in full before quoting their numbers.

Free copies, no login needed:

- Paper 1: NIOSH copy
- Paper 2: Lafayette College copy
- Paper 3: NIOSH publication 2007-146
- Paper 7: Eric Marks' PhD thesis, Georgia Tech (Chapter 8 covers the same test method)
- Paper 8: NIOSH copy
- Paper 9: MDPI Sensors
- Paper 10: IEEE Access, open access
- Paper 12: NIOSH copy
- Paper 15: CVF open-access PDF

Other pages used:

- OpenAlex record for LANDMARC (citation count)
- Seeing Through Fog project page, Princeton
- UWB LOS/NLOS dataset for paper 10 and its code